

MULTIMEDIA



UNIVERSITY

STUDENT ID NO

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MULTIMEDIA UNIVERSITY

FINAL EXAMINATION

TRIMESTER 1, 2016/2017

TCN2141 – COMPUTER NETWORKS
(All sections / Groups)

19 OCTOBER 2016

2.30 p.m. – 4.30 p.m.

(2 Hours)

INSTRUCTIONS TO STUDENTS

1. This Question paper consists of 4 pages with 6 Questions only
2. Attempt **ALL SIX** questions.
3. All questions carry equal marks and the distribution of the marks for each question is given.
4. Please print all your answers in the Answer Booklet provided.

QUESTION 1

- a) Given a host IP address with CIDR as 177.32.129.2 /20. As the network administrator, find the following information?
- What is the given current IP address subnet range?
 - What is the subnet number for the 5th subnet?
 - What is the subnet broadcast address for 3rd subnet?
 - What are assignable addresses for the 8th subnet?
 - How many usable hosts per subnet?
- b) Describe the difference between transmission delay, queuing delay, and propagation delay?
- c) What is three-way handshake? Why virtual circuit approach requires three way handshake process?

[5 + 3 + 2 = 10 Marks]

QUESTION 2

- a) Consider sending a 5700-byte datagram into a link that has a MTU of 1300 bytes. Suppose the original datagram is stamped with the identification number 111516 as shown in table below (Original IP datagram). How many fragments are generated? What are the values in the various fields in the IP datagram(s) generated related to fragmentation? Use IP fragments table below to give your answer.

Original IP Datagram							
Sequence	Identifier	Total Length	DF	MF	Offset		
AB	111516	5700	0	0	0		
IP Fragments (Ethernet)							
Sequence	Identifier	Total Length	DF	MF	Offset		

Continued...

- b) State the FOUR tasks that must be performed by a router in link state routing to ensure that each node has the routing table showing the least-cost node to every other node?
- c) Briefly explain the differences between Circuit Switching and Virtual Circuit Packet Switching.

[5 + 4 + 1 = 10 Marks]

QUESTION 3

- a) What are the TWO basic approaches used to perform a distributed route computation, and how does each work?
- b) What is a routing loop?
- c) Briefly describe Split Horizon Technique.
- d) What is the Poison reverse technique?

[4 + 2 + 2 + 2 = 10 Marks]

QUESTION 4

- a) A multicast address for a group is 231.185.58.10. What is its 48-bit Ethernet address for a LAN using TCP/IP?
- b) Congestion control provides mechanism that can prevent congestion before it happens. Briefly describe FOUR open-loop congestion control policies.
- c) IPSec is an open standard framework for ensuring secure private communication over IP networks. Compare TWO protocols used to provide data confidentiality, authenticity, integrity, and replay protection.

[3 + 4 + 3 = 10 Marks]

Continued...

QUESTION 5

- a) The following is a dump of a TCP header in hexadecimal format.

04320013 00000021 00000000 50020AB5 00000000

- i. What is the source port number?
 - ii. What is the destination port number?
 - iii. What the sequence number?
 - iv. What is the acknowledgment number?
 - v. What is the length of the header?
 - vi. What is the type of the segment?
 - vii. What is the window size?
 - viii. What is checksum value?
- b) List the THREE IPv6 address types, and give a brief explanation of each.
- c) What is the role of Network Management System (NMS) of SNMP? Give TWO advantages of implementing SNMP function in network.

[4 + 3 + 3 = 10 Marks]

QUESTION 6

- a) What are the differences between a classful addressing and classless addressing in IPv4?
- b) What is the purpose of a default mask in IPv4 addressing?
- c) Find the class of the following IP addresses.
- i. 11000001.00101100.00110110.00000001
 - ii. 11101101.00001110.00000001.00000110
 - iii. 108.64.3.7
 - iv. 129.54.16.8
- d) Give the shorter version of the IPv6 address?
- i. 2001:1934:0101:0000:0000:0000:0000:0035
 - ii. 2001:A0D3:0000:0000:0343:0000:0000:0323
- e) Explain Network Address Translation (NAT) and state one of its benefits

[2+2+2+2+2 = 10 Marks]

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